Casey Barton

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PROFILE

Applied-AI systems engineer based in Honolulu. I architect and build multi-agent workflows for code generation, testing, and review — moving work from ticket to PR with minimal friction — grounded in RAG-driven context and CI/CD reality, not demos. My systems ship with observability and the eval frameworks that make non-deterministic systems shippable: managing hallucinations, designing feedback loops, and measuring quality before and after every change.

Three years running an evidence-grade agent control plane: typed state transitions, approval gates, audit receipts, recoverable runs. Backend/platform engineering across a 1,413-repository estate — **65 MCP connectors**, **989-test** suite, **95/95** verification checks green. I understand why code fails, how to secure a pipeline, and how to build for observability.

CORE COMPETENCIES

Engineering: Python · TypeScript · Node.js · SQL · Bash · REST APIs · Supabase · Postgres · Vercel · GitHub Actions (CI/CD) · containers

AI & Agents: multi-agent workflows (code gen, testing, review, static analysis) · RAG pipelines & RAG-driven context · LLM evaluation frameworks · prompt & context engineering · MCP protocol · function calling & structured outputs

Production: CI/CD integration · observability & run logging · fail-closed recovery · root-cause quarantine · feedback loops for non-deterministic systems · idempotent automation

EXPERIENCE

Founder / Applied AI Systems Architect 2023 – Present

GlacierEQ — Honolulu, HI

- Own the loop: hand-rolled multi-agent loops with function calling, structured outputs, and typed handoff contracts — agentic workflows that move work from intake to verified output with minimal friction, integrated with CI/CD (GitHub Actions) as the delivery reality.
- Evaluation-first delivery (job-app-helix): **989-test** agent control plane; robust evaluation frameworks for non-deterministic systems — gold sets, LLM-as-judge with written rubrics, replay loops — the feedback loops that manage hallucinations and prove quality instead of asserting it.
- Stateful retrieval (unified-memory-mcp, mcp-iceberg-mansion): Mem0, Supermemory, Supabase pgvector, Pinecone, and Neo4j federated behind one MCP interface — parallel fan-out, content-hash dedup; the RAG-driven context that grounds agents in real code and docs.
- Multi-agent orchestration (anthropic-agent-coordinator): deterministic DAG scheduler, **95/95** tests green; tool calls proposed for review, never executed blind — custom-built state machines where the right tool beats the framework.
- Production reliability (apex-control-plane): SHA-256 transport envelopes, idempotency keys, circuit breakers, dead-letter capture, fail-closed startup; failure analysis with root-cause quarantine instead of silent retries — why code fails, instrumented.

- Operate a **1,413-repository** applied-AI estate — **65 MCP connectors** integrated with claim tracking and idempotent automation; **20+ production repositories** maintained under test and CI; iterate in production, measured by reduced cycle time and cognitive load.

Chief Inspector 2024

Diamond Head Home Inspections — Honolulu, HI

- Led residential inspection operations: field assessments, defect documentation, and client reporting — direct customer-facing technical work; process judgment from regulated, evidence-bound work.

EDUCATION

University of Hawaii at Manoa 2016

Bachelor of Science

AWS Cloud Institute 2025

Cloud Computing program (brief attendance)

Colorado State University

Undergraduate coursework

SELECTED PROOF

Job-App Helix — attributable campaign engine

- Attributable campaign engine with immutable evidence models, stable finding codes, and fail-closed **GO / NO-GO** decisions — the eval gate that decides whether an agent change ships.
- Verification gates on every agent action: no agent completion claim without independent provider readback.

GlacierEQ estate — agent systems portfolio

- Trace intelligence wired into every run: full logging of prompts, tool calls, and decisions — any execution reconstructable and debuggable from its trace; failed evals diagnosed from evidence.
- Per-task latency instrumentation and decision audit logs across agent runs — the tracing practice that makes benchmarking honest.

SKILLS

Languages: Python · TypeScript · SQL · Bash · JavaScript / Node.js

Applied AI: multi-agent workflows · RAG & retrieval-grounded generation · LLM evaluation frameworks · prompt & context engineering · MCP protocol

Data & Infra: Supabase · Postgres · Vercel · GitHub Actions (CI/CD) · REST APIs · containers

Practice: observability · feedback loops · fail-closed recovery · idempotent automation · clear human-facing documentation

Evidence boundary: portfolio repositories are independent work. Domain-inspired demonstrations are not claims of company affiliation, production deployment at eMAX, healthcare-domain production depth, or certification.